Representation Learning

1 Dataset Understanding

Fashion-MNIST dataset consists of 28×28 grayscale images and 10 clothing categories. From visual checking it is seen that the main variation in images are shape, structure, and texture.

Clothing categories are mainly differed by their overall shape. For example, trousers are vertically elongated, shirts have sleeves, and shoes have curved structures.

- The most important features for classification are the overall shape and structure of the clothing.
- small pixel-level details like textures or minor noise don't seem to play a major role.

2 Model Setup

A fully connected autoencoder was implemented. Three models were trained with different latent dimensions:

- Small (latent = 4): Strong compression
- Medium (latent = 24): Balanced representation
- Large (latent = 128): High capacity

All other parameters were kept constant to only see effect of latent size

3 Training Behavior

- The reconstruction loss decreased steadily for all models.
- As expected, the large model achieved the lowest loss, followed by the medium and then the small model.

but, this difference in loss does not directly indicate better learning, but rather shows the model's capacity to store information.

4 Reconstruction Analysis

- The small model produces blurry outputs where fine details are lost.
- the general structure of the objects is still recognizable.
- This shows that the model retains only the most essential features.

The medium model got a better balance. It preserves both structure and some level of detail, producing better reconstructions.

- The large model reconstructs images preserving even minor details.
- but, some reconstructions, while visually sharp, feel slightly unnatural or overly smoothed, showing that the model may be memorizing patterns rather than learning meaningful data

important observation: good reconstruction quality does not necessarily mean good representation learning.

5 Latent Space Analysis

Latent representations were visualized using PCA.

- For the small model, clusters corresponding to different classes are relatively well-separated.
- This shows that compression forces the model to organize the data in a meaningful way.
- The medium model shows the clearest separation between classes, suggesting that it learns the most useful representation among the three.
- the large model shows more overlap between classes.
- The latent space appears less structured, likely because the model is storing unnecessary details instead of organizing information categorically.

Overall, the medium model provides the best trade-off between separation and structure.

6 Interpolation Analysis

Interpolation between two different images shows us how smooth the latent space is.

- In the small model, transitions are smooth but somewhat blurry.
- The transformation from one class to another is gradual, indicating a structured latent space.
- The medium model produces the most natural transitions.
- Intermediate images blend characteristics of both classes in the best way.
- The large mode is detailed,but shows less meaningful transitions.
- Some intermediate images appear inconsistent or overly noisy, indicating that the latent space lacks proper organization.

important observation: well-structured latent space is more important than simply preserving detail.

7 Capacity and Failure Modes

- The first information to be lost in smaller models is fine texture and pixel-level detail.
- Larger models begin to memorize data rather than extract meaningful features.
- The latent space becomes less interpretable as capacity increases.

The small model fails by losing too much detail, while the large model fails by losing meaningful structure. The medium model shows a good balance between these extremes.

8 Final Analysis

Preservation: All models preserve the overall shape and structure of clothing items. This shows to be the most important feature for representation.

Loss: Fine details such as textures and small variations are lost in smaller latent spaces. This indicates that such information is not important for reconstruction under compression.

Entanglement: In smaller models, multiple features such as shape and class identity become entangled when you limit dimensions. This can make separation harder but also forces meaningful encoding.

Capacity Effects: Increasing latent size improves reconstruction quality but reduces the meaningful organization of latent space. Excess capacity leads to memorization.

Downstream Utility: smaller and medium models are likely more useful for downstream tasks, as they capture essential features while ignoring noise. Large models, despite low loss, may not generalize well because they memorize more.

9 Conclusion

This sshows that representation learning is mainly about balancing compression and information retention. Smaller latent spaces force the model to learn meaningful abstractions, while larger ones allow memorization.

The results clearly show that lower reconstruction loss does not imply better learning. Instead, the quality of the latent space and its structure determine how useful the learned representation truly is.